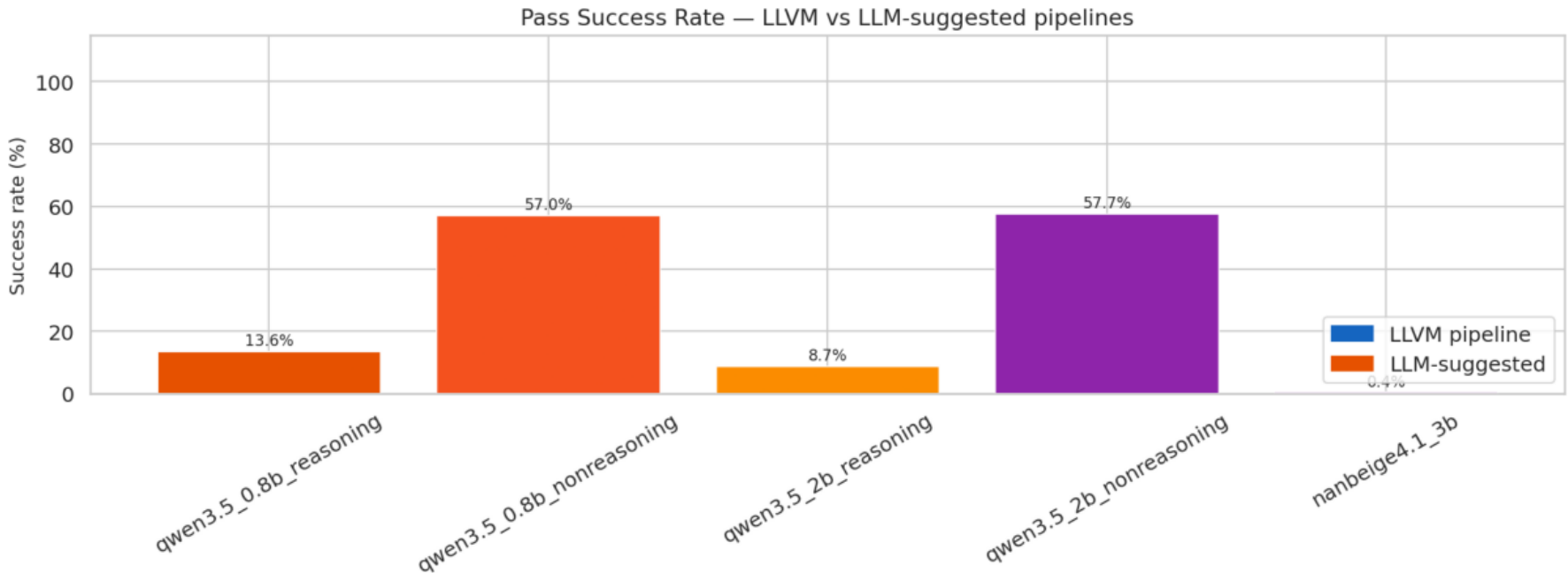
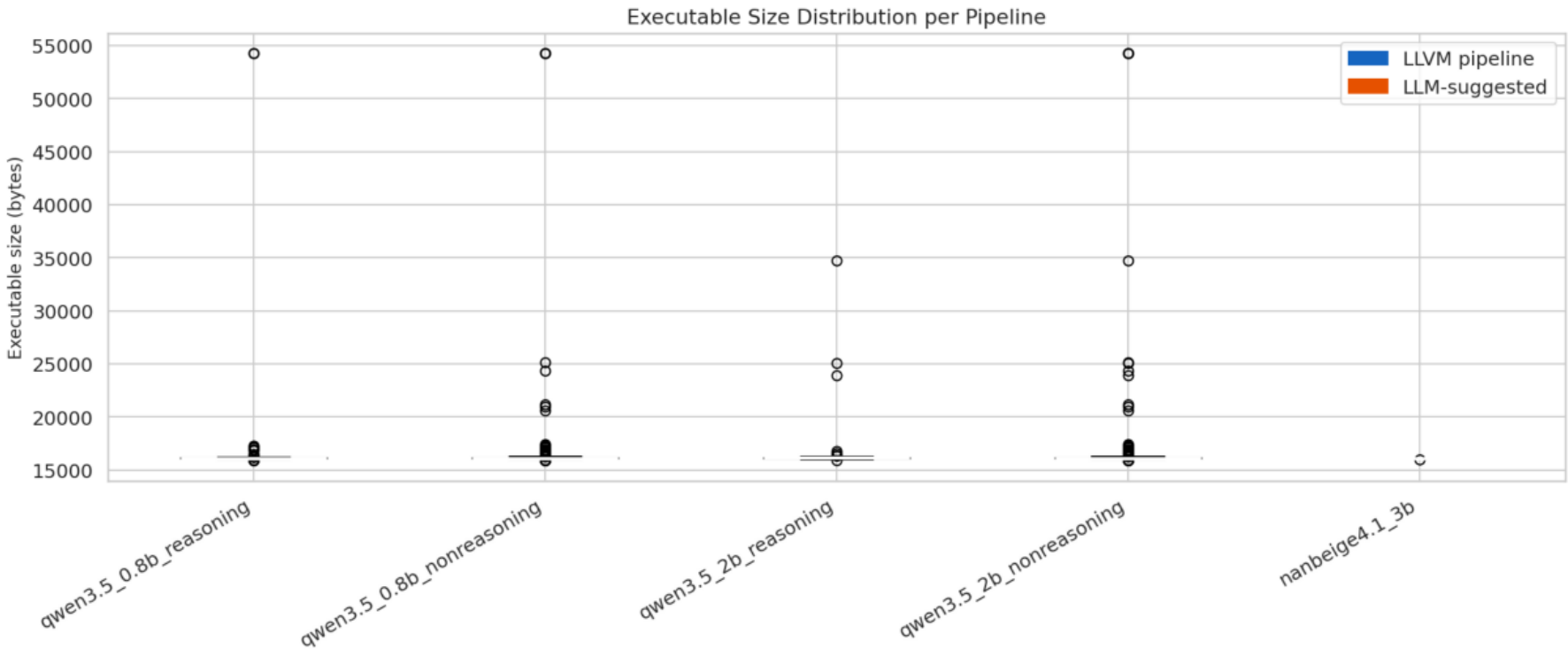
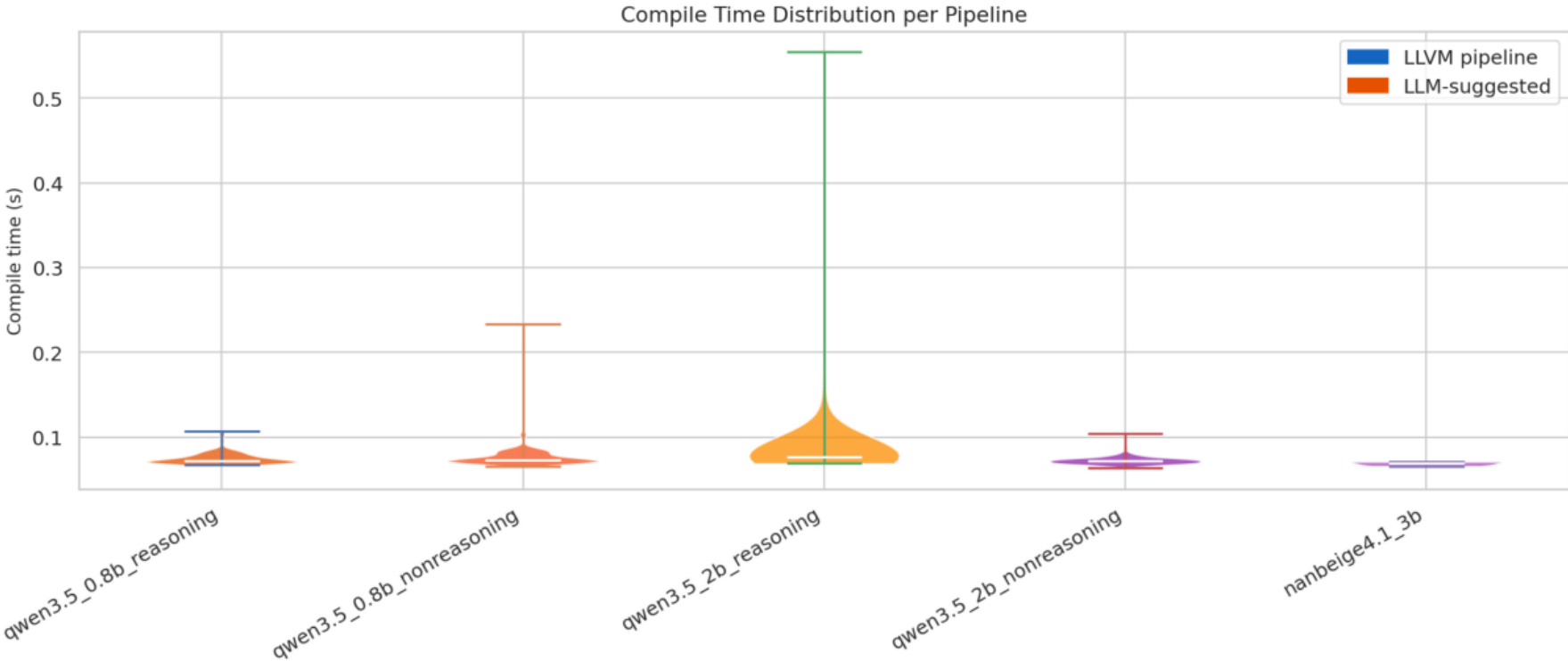
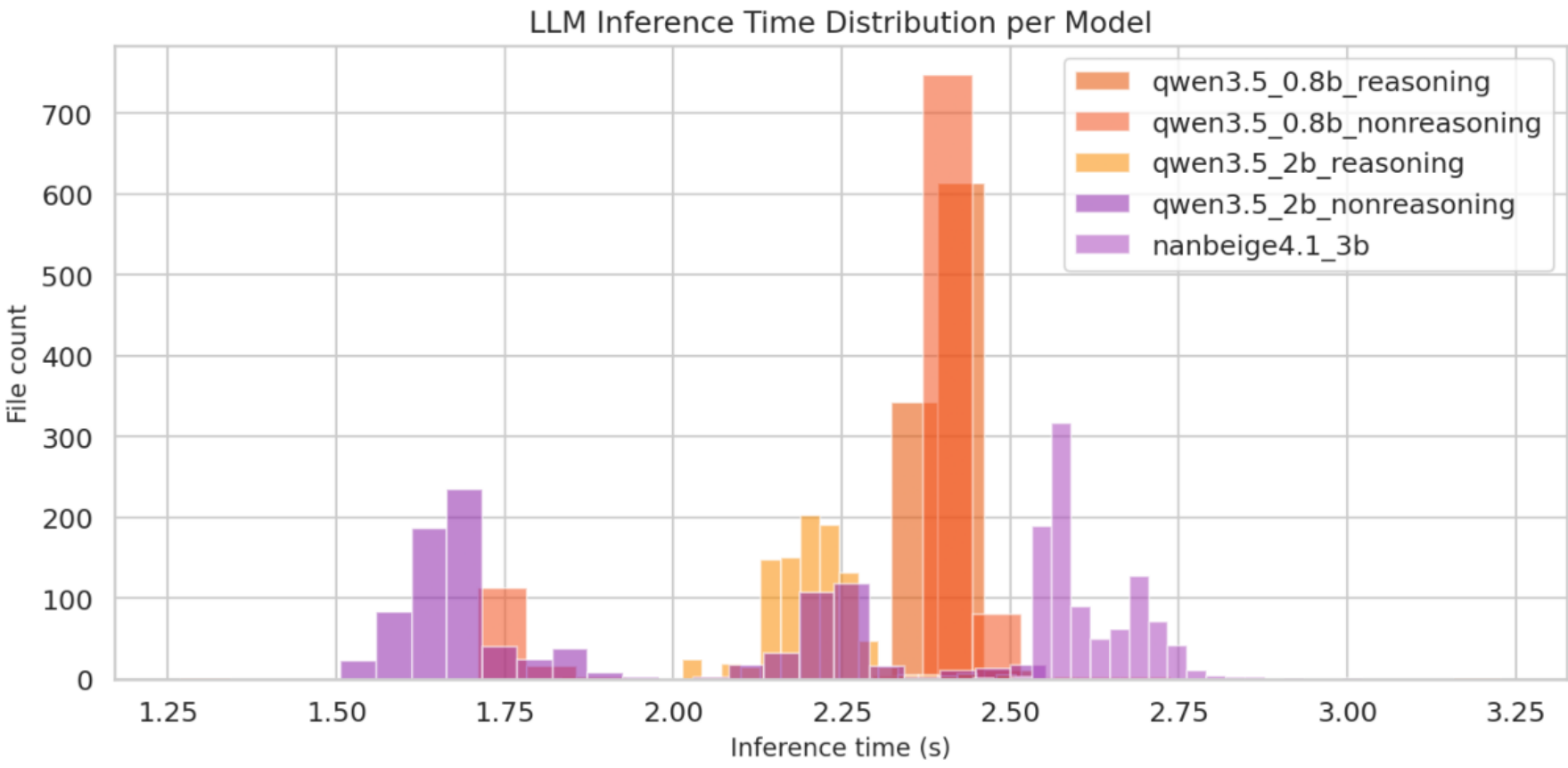
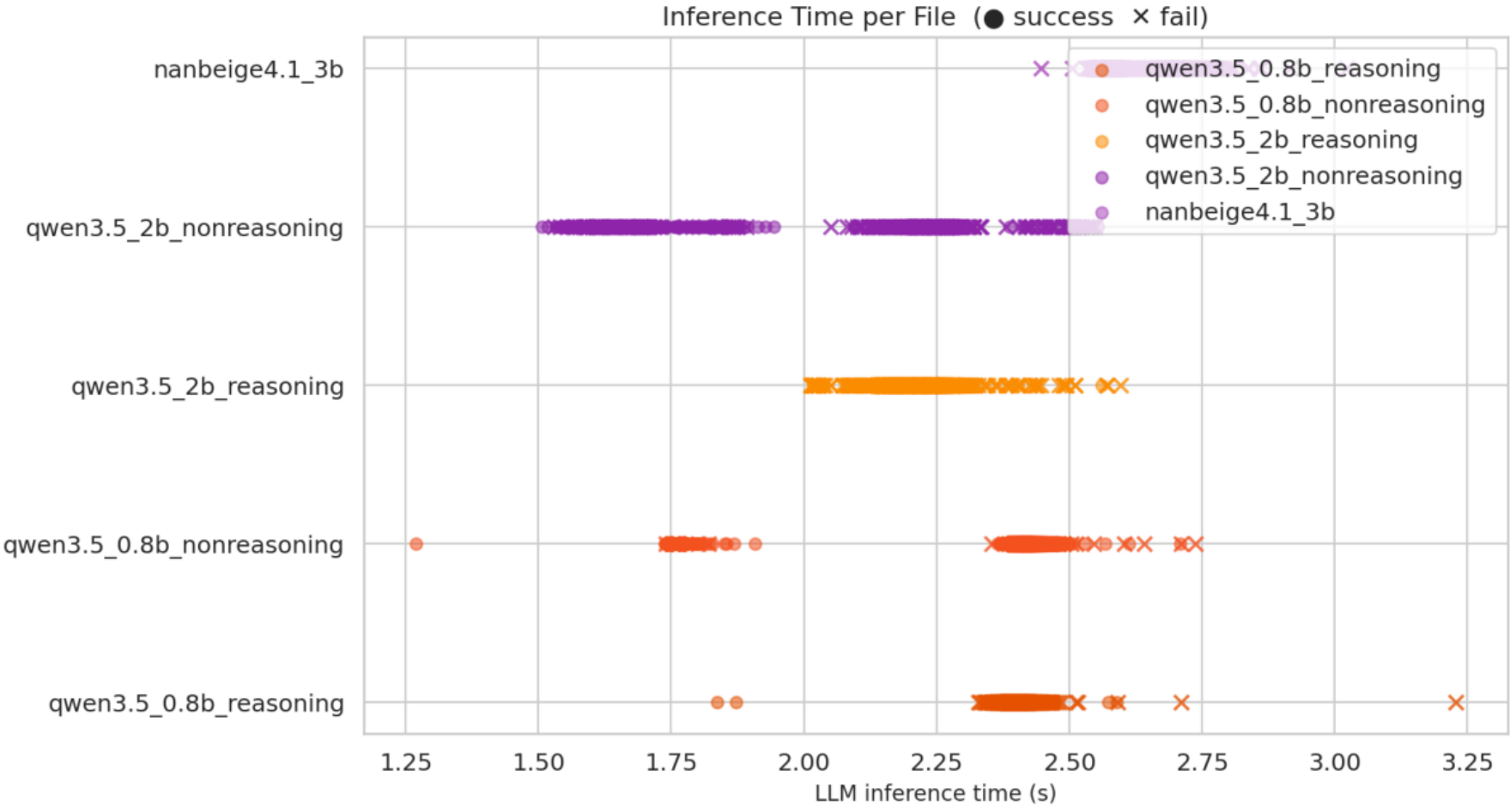
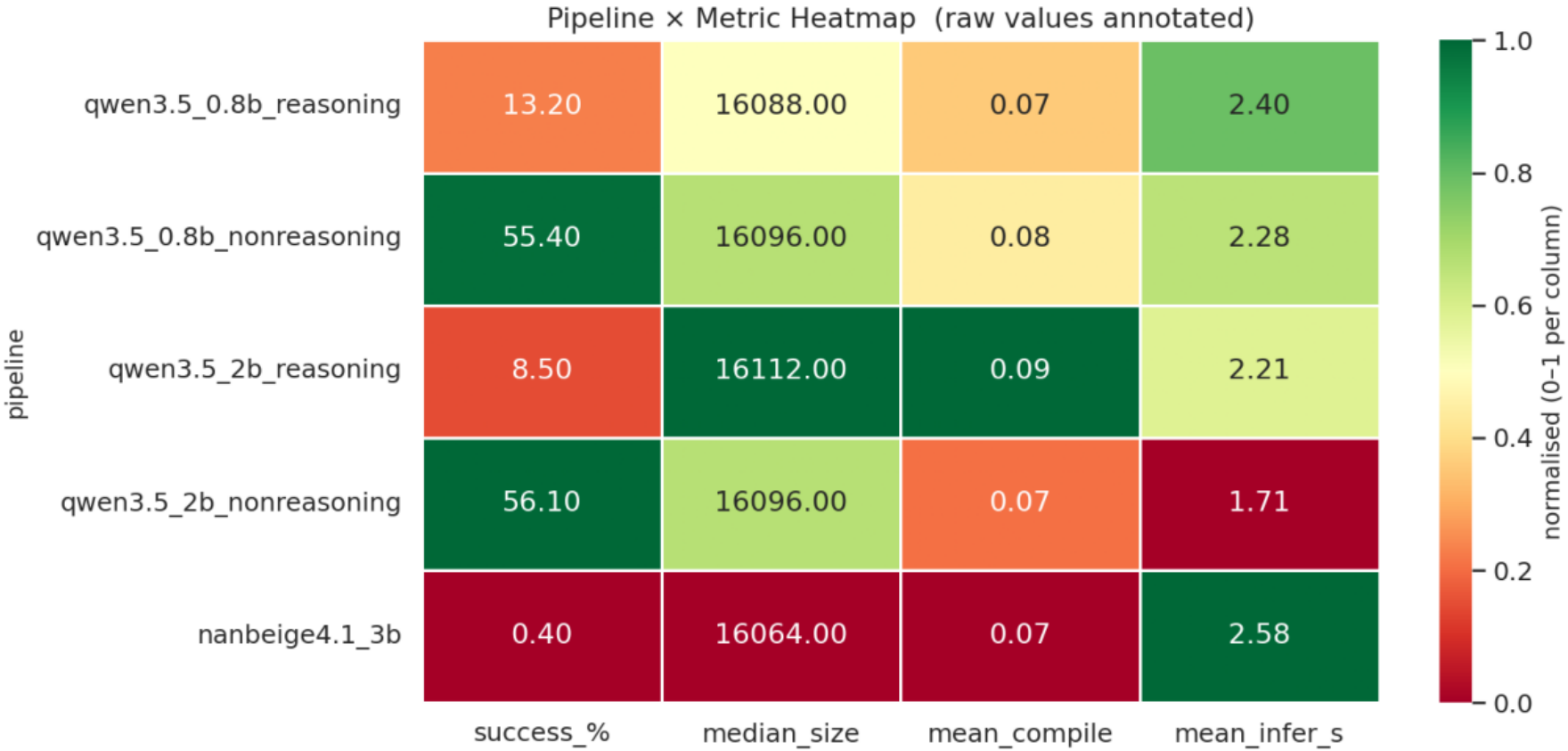

Pass Success Rate — LLVM vs LLM-suggested pipelines
Success rate (%)
100
80
60
40
20
0
13.6%
57.0%
8.7%
57.7%
0.4%
LLVM pipeline
LLM-suggested
qwen3.5_0.8b_reasoning
qwen3.5_0.8b_nonreasoning
qwen3.5_2b_reasoning
qwen3.5_2b_nonreasoning
nanbeige4.1_3b


Executable Size Distribution per Pipeline
Executable size (bytes)
55000
50000
45000
40000
35000
30000
25000
20000
15000
LLVM pipeline
LLM-suggested
qwen3.5_0.8b_reasoning
qwen3.5_0.8b_nonreasoning
qwen3.5_2b_reasoning
qwen3.5_2b_nonreasoning
nanbeige4.1_3b


Compile Time Distribution per Pipeline
LLVM pipeline
LLM-suggested
Compile time (s)
0.5
0.4
0.3
0.2
0.1
qwen3.5_0.8b_reasoning
qwen3.5_0.8b_nonreasoning
qwen3.5_2b_reasoning
qwen3.5_2b_nonreasoning
nanbeige4.1_3b


LLM Inference Time Distribution per Model
qwen3.5_0.8b_reasoning
qwen3.5_0.8b_nonreasoning
qwen3.5_2b_reasoning
qwen3.5_2b_nonreasoning
nanbeige4.1_3b
File count
0
100
200
300
400
500
600
700
1.25
1.50
1.75
2.00
2.25
2.50
2.75
3.00
3.25
Inference time (s)


Inference Time per File (● success × fail)
nanbeige4.1_3b
qwen3.5_2b_nonreasoning
qwen3.5_2b_reasoning
qwen3.5_0.8b_nonreasoning
qwen3.5_0.8b_reasoning
qwen3.5_0.8b_reasoning
qwen3.5_0.8b_nonreasoning
qwen3.5_2b_reasoning
qwen3.5_2b_nonreasoning
nanbeige4.1_3b
1.25
1.50
1.75
2.00
2.25
2.50
2.75
3.00
3.25
LLM inference time (s)


Pipeline × Metric Heatmap (raw values annotated)
pipeline
qwen3.5_0.8b_reasoning
qwen3.5_0.8b_nonreasoning
qwen3.5_2b_reasoning
qwen3.5_2b_nonreasoning
nanbeige4.1_3b
13.20
16088.00
0.07
2.40
55.40
16096.00
0.08
2.28
8.50
16112.00
0.09
2.21
56.10
16096.00
0.07
1.71
0.40
16064.00
0.07
2.58
success_%
median_size
mean_compile
mean_infer_s
normalised (0–1 per column)
1.0
0.8
0.6
0.4
0.2
0.0

No LLVM baseline data available